Week 2 Assignment Solutions

2. Word Reversal Encoder

Reverse each word individually while keeping the original word order. The input is assumed to use single spaces between words.

Java Solution

```
public class Main {
    static String reverseEachWord(String sentence) {
        String[] words = sentence.split(" ");
        StringBuilder result = new StringBuilder();

        for (int i = 0; i < words.length; i++) {
            StringBuilder reversed = new StringBuilder();
            for (int j = words[i].length() - 1; j >= 0; j--) {
                reversed.append(words[i].charAt(j));
            }
            if (i > 0) {
                result.append(" ");
            }
            result.append(reversed);
        }
        return result.toString();
    }

    public static void main(String[] args) {
        System.out.println(reverseEachWord("hello club"));
    }
}
```

Sample Output

```
olleh bulc
```

Notes

The nested loops build each word in reverse, then append the reversed words to one StringBuilder separated by spaces.